

Bachelor AI - MLOps

05A INFRASTRUCTURE OVERVIEW

Tobias Mérinat

`tobias.merinat@hslu.ch`

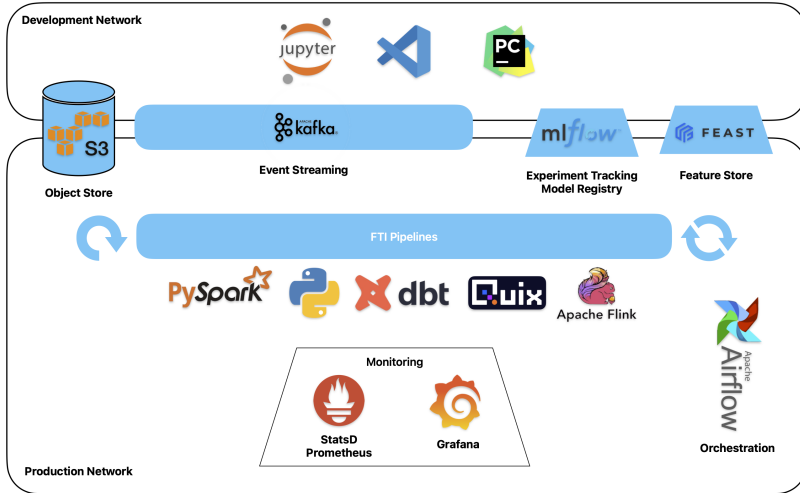


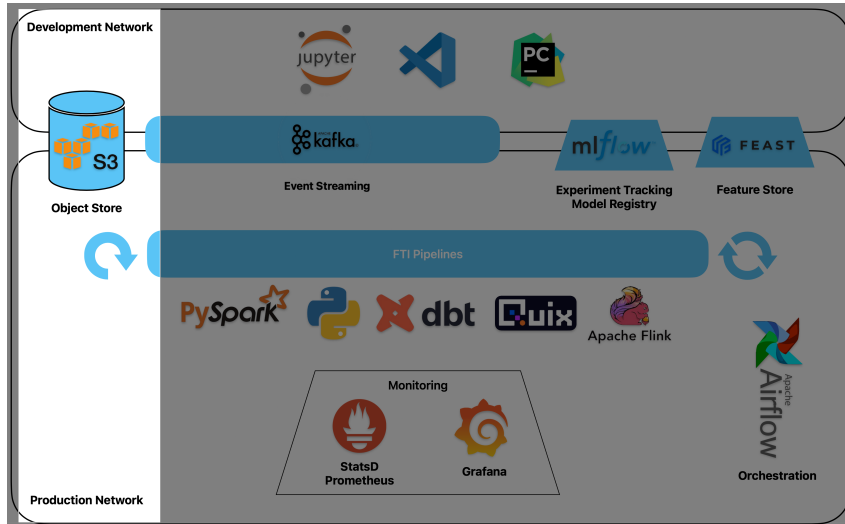
DEPARTMENT OF INFORMATION TECHNOLOGY
Lucerne University of Applied Sciences and Arts
6343 Rotkreuz, Switzerland

September 17/24 and October 15 2026

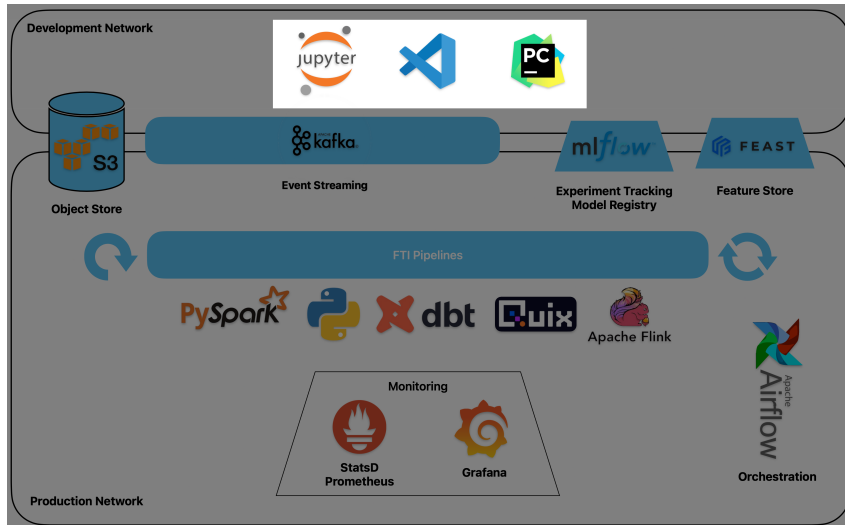
- ML Models live inbetween the responsibilities of
 - Data scientists (who build the models)
 - Data engineers (who build the data pipelines and use the models)
 - Operations engineers (who build and run the infrastructure)
- We need to understand the responsibilities and tools these groups use
 - As a data scientist, you need to be aware how the infrastructure works
 - As a data engineer, you need a basic understanding of Machine Learning
 - As a operations engineer, you need to know how models work and how to optimize them

Infrastructure Big Picture

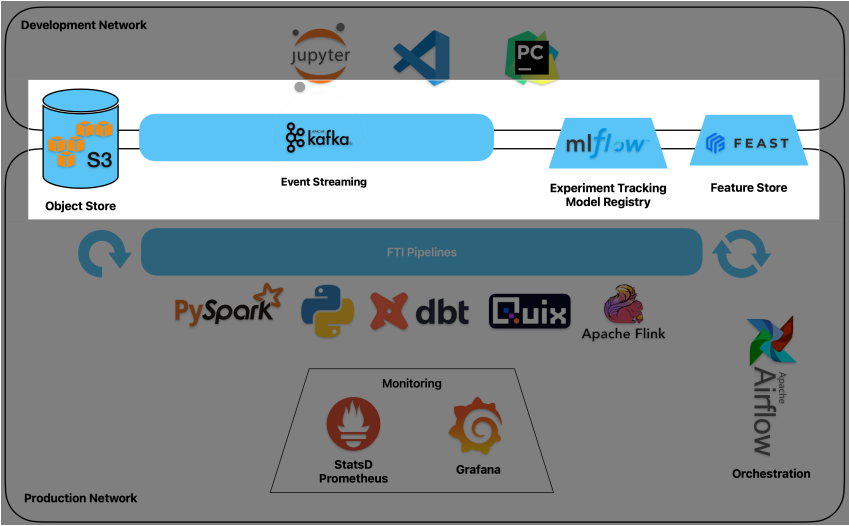


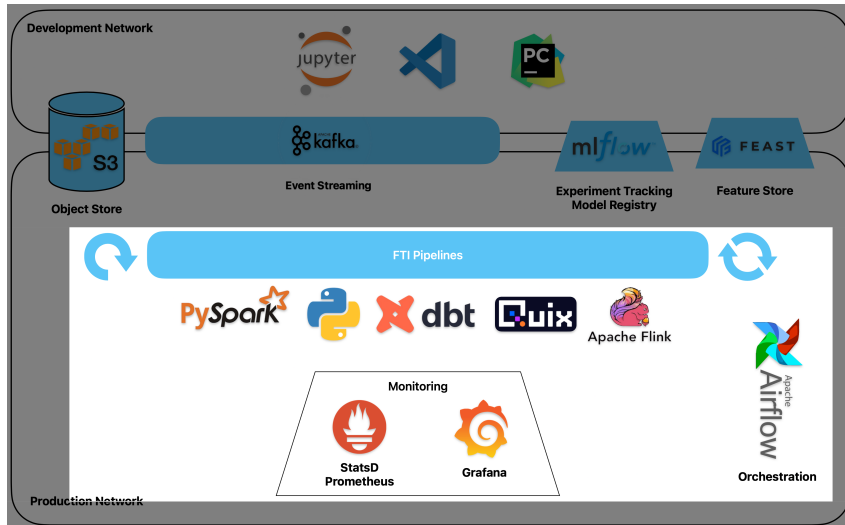


Development

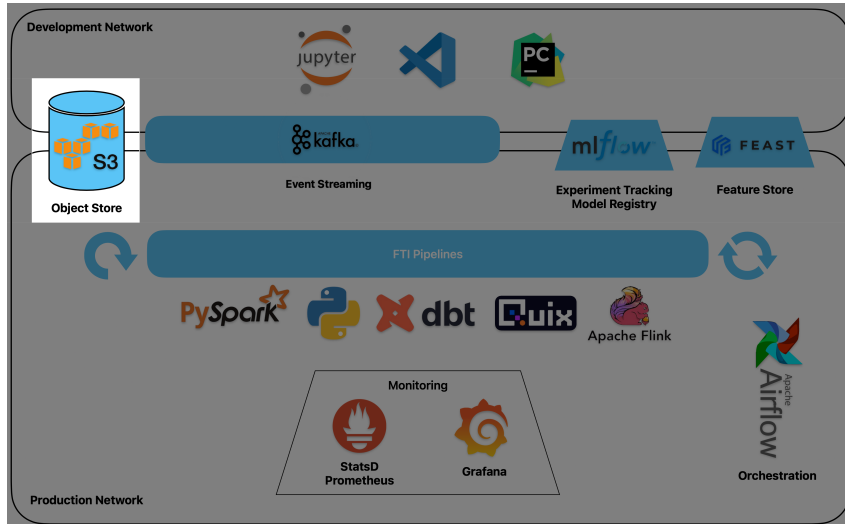


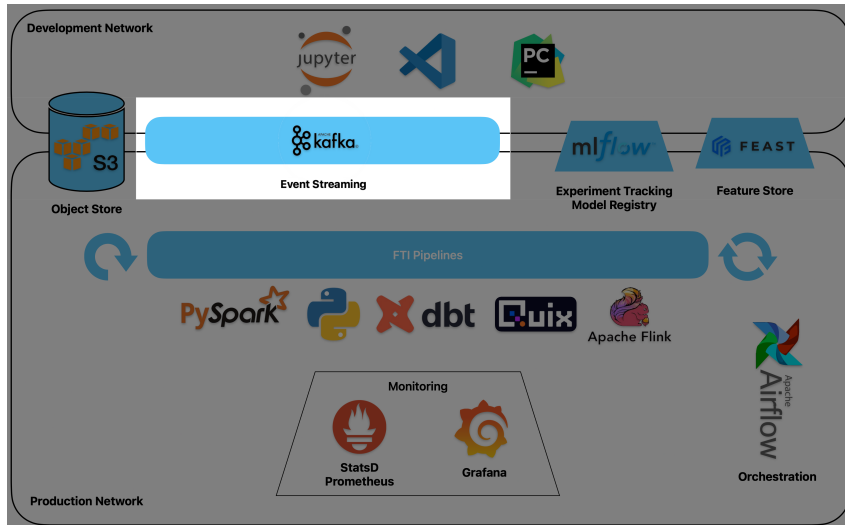
Shared Infrastructure



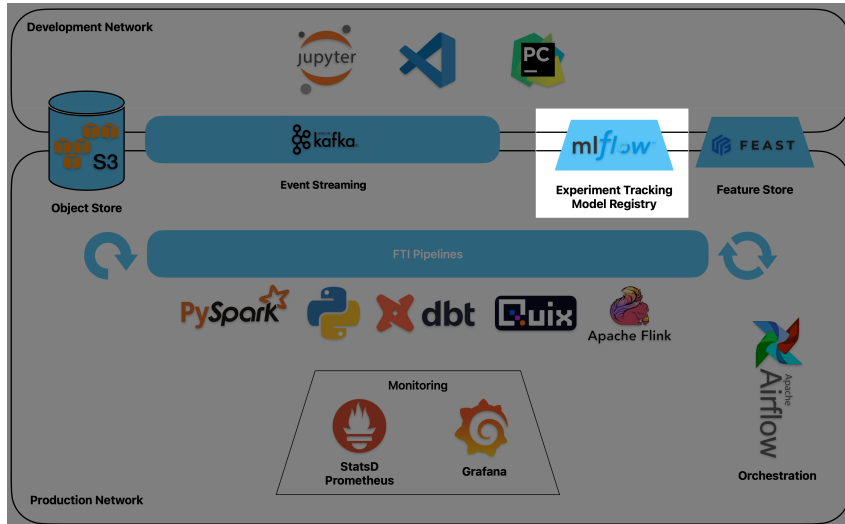


Object Store

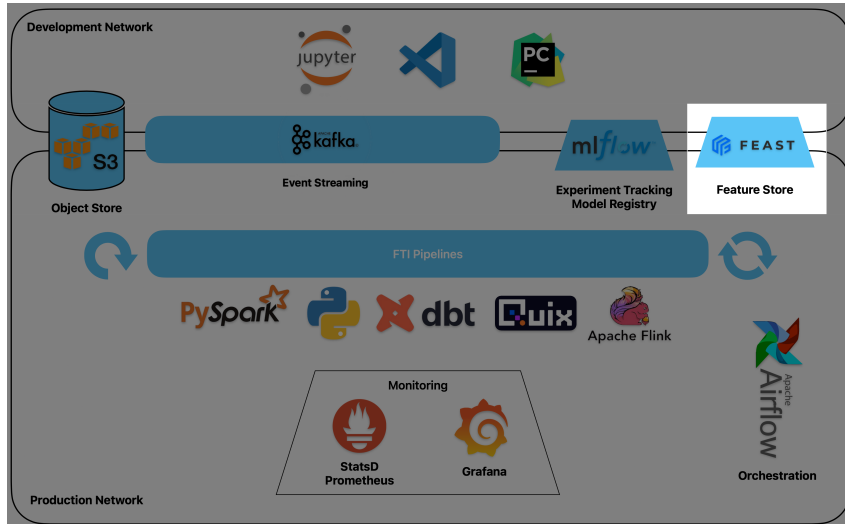




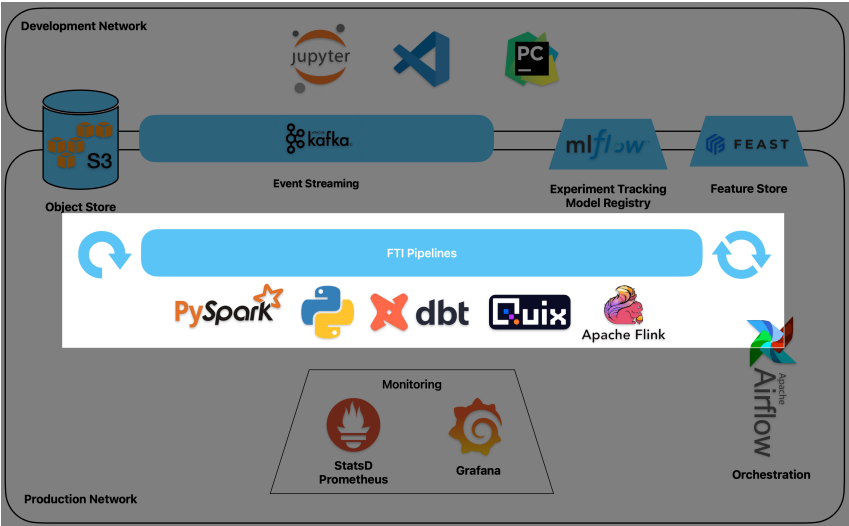
Model Registry and Experiment Tracking



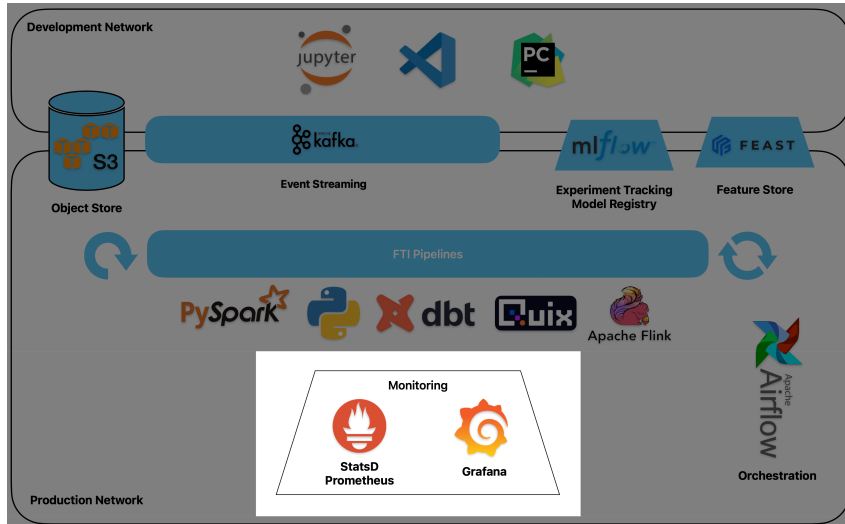
Feature Store



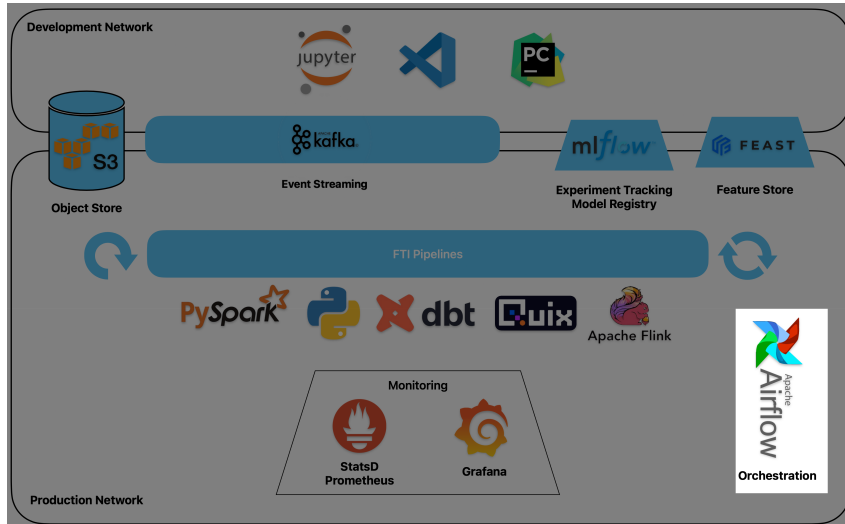
Batch and Streaming Pipelines



Monitoring



Orchestration



Overview again

